

Learning Outcome(s) Assessed

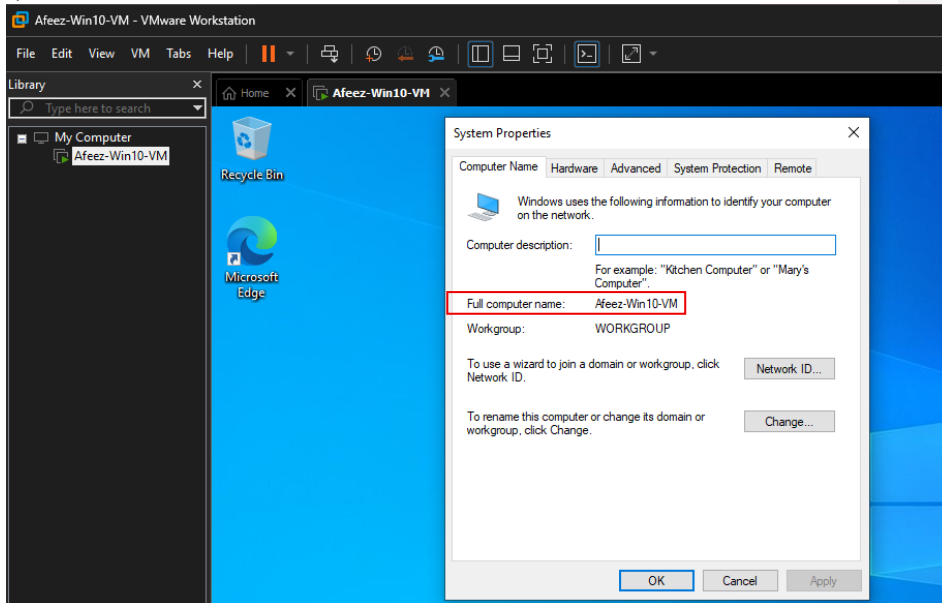
- Configure end-point solutions for end user access using a predetermined process
- Automate repetitive configuration tasks using scripting

Overview

In this assignment you're going to create a script to bulk install 3 applications silently. Then you are going to prepare a Win 10/11 VM as a "golden image" to be installed on another machine. Finally you are going to pair up and install another student's "golden image" and give them feedback via a provided checklist.

Part A: Sssh! Install these applications: (Week 12)

- 1) Create a Win10/11 VM, the specs should match the "Minimum Recommended" from Microsoft. Ensure the Computer Name is your own. Submit a screenshot that includes the Computer Name.



- 2) Research, create and use a script to automate the silent installation of 3 applications they can be installed in any order. For example: Google Chrome, NotePad++, and one more.

Provide a text copy (.txt) of your script you used to install the 3 applications. Include all sources used for how you created script.

Commented [PS1]: APA? How about just a link?

```

# -----
# Silent install script for Chrome, Notepad++, and Git
# Run PowerShell as Administrator if possible
# -----

# Create installer folder on Desktop
$InstallerPath = "$env:USERPROFILE\Desktop\Installers"
if (-Not (Test-Path $InstallerPath)) {
    New-Item -ItemType Directory -Path $InstallerPath | Out-Null
}

# Installer URLs
$ChromeURL =
"https://dl.google.com/dl/chrome/install/googlechromestandaloneenterprise64.msi"
$NotepadURL = "https://github.com/notepad-plus-plus/notepad-plus-
plus/releases/download/v8.6.7/npp.8.6.7.Installer.x64.exe"
$GitURL = "https://github.com/git-for-
windows/git/releases/download/v2.47.0.windows.1/Git-2.47.0-64-bit.exe"

# File paths
$ChromeInstaller = Join-Path $InstallerPath "Chrome.msi"
$NotepadInstaller = Join-Path $InstallerPath "NotepadPP.exe"
$GitInstaller = Join-Path $InstallerPath "Git.exe"

# Function to download files
function Download-File($url, $destination) {
    Write-Output "Downloading $destination ..."
    $wc = New-Object System.Net.WebClient
    $wc.DownloadFile($url, $destination)
    Write-Output "Downloaded $destination"
}

# Download installers
Download-File $ChromeURL $ChromeInstaller
Download-File $NotepadURL $NotepadInstaller
Download-File $GitURL $GitInstaller

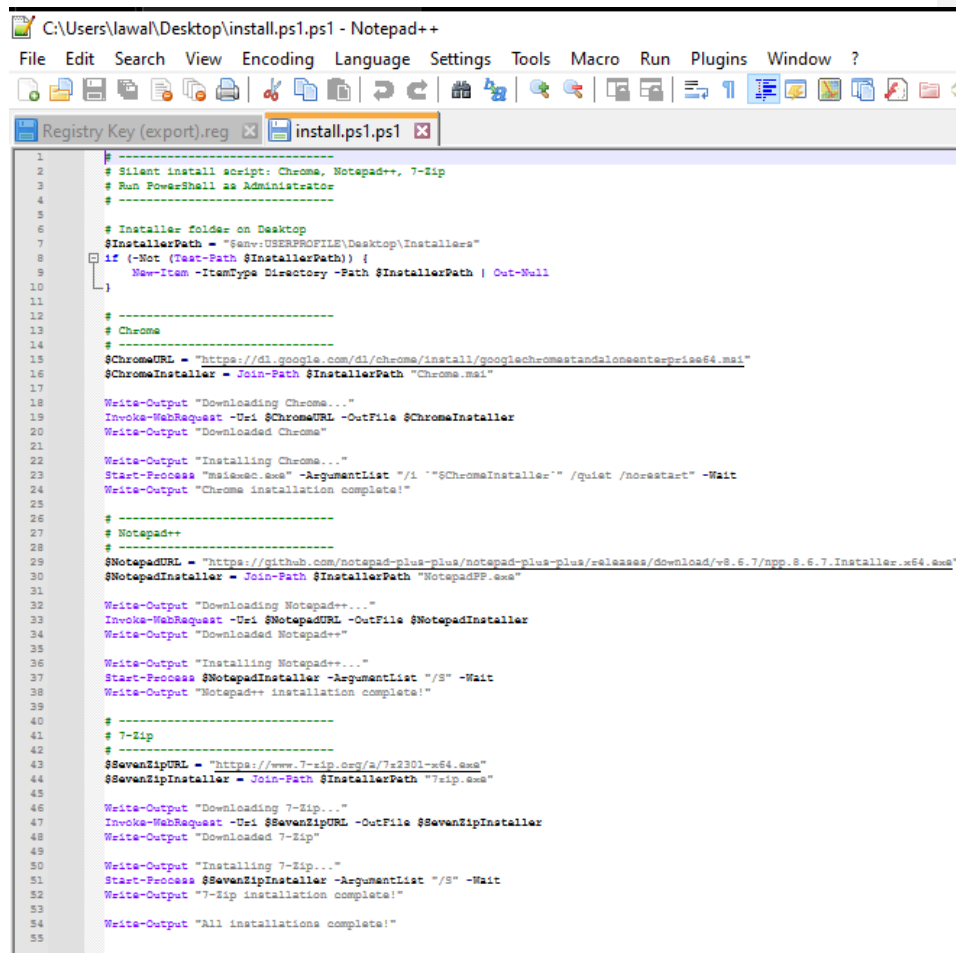
# Install Chrome silently
Write-Output "Installing Chrome..."
Start-Process "msiexec.exe" -ArgumentList "/i `"$ChromeInstaller`" /quiet /norestart" -Wait

```

```
# Install Notepad++ silently
Write-Output "Installing Notepad++..."
Start-Process $NotepadInstaller -ArgumentList "/S" -Wait
```

```
# Install Git silently
Write-Output "Installing Git..."
Start-Process $GitInstaller -ArgumentList "/VERYSILENT /NORESTART" -Wait
```

```
Write-Output "All installations complete."
```



```
1  # -----
2  # Silent install script: Chrome, Notepad++, 7-Zip
3  # Run PowerShell as Administrator
4  # -----
5
6  # Installer folder on Desktop
7  $InstallerPath = "$env:USERPROFILE\Desktop\Installers"
8  if (-Not (Test-Path $InstallerPath)) {
9      New-Item -ItemType Directory -Path $InstallerPath | Out-Null
10 }
11
12 # -----
13 # Chrome
14 # -----
15 $ChromeURL = "https://dl.google.com/dl/chrome/install/googlechromestandaloneenterprise64.msi"
16 $ChromeInstaller = Join-Path $InstallerPath "Chrome.msi"
17
18 Write-Output "Downloading Chrome..."
19 Invoke-WebRequest -Uri $ChromeURL -OutFile $ChromeInstaller
20 Write-Output "Downloaded Chrome"
21
22 Write-Output "Installing Chrome..."
23 Start-Process "msiexec.exe" -ArgumentList "/i \"$ChromeInstaller\" /quiet /norestart" -Wait
24 Write-Output "Chrome installation complete!"
25
26 # -----
27 # Notepad++
28 # -----
29 $NotepadURL = "https://github.com/notepad-plus-plus/notepad-plus-plus/releases/download/v8.6.7/npp.8.6.7.Installer.msi"
30 $NotepadInstaller = Join-Path $InstallerPath "NotepadPP.msi"
31
32 Write-Output "Downloading Notepad++..."
33 Invoke-WebRequest -Uri $NotepadURL -OutFile $NotepadInstaller
34 Write-Output "Downloaded Notepad++"
35
36 Write-Output "Installing Notepad++..."
37 Start-Process $NotepadInstaller -ArgumentList "/S" -Wait
38 Write-Output "Notepad++ installation complete!"
39
40 # -----
41 # 7-Zip
42 # -----
43 $SevenZipURL = "https://www.7-zip.org/a/7z2301-m64.exe"
44 $SevenZipInstaller = Join-Path $InstallerPath "7zip.exe"
45
46 Write-Output "Downloading 7-Zip..."
47 Invoke-WebRequest -Uri $SevenZipURL -OutFile $SevenZipInstaller
48 Write-Output "Downloaded 7-Zip"
49
50 Write-Output "Installing 7-Zip..."
51 Start-Process $SevenZipInstaller -ArgumentList "/S" -Wait
52 Write-Output "7-Zip installation complete!"
53
54 Write-Output "All installations complete!"
55
```

Sources:

- Google. (n.d.). *Deploy Chrome Enterprise*. Google Chrome Enterprise Help. Retrieved December 1, 2025, from <https://support.google.com/chrome/a/answer/9915669?hl=en>
- Notepad++ Team. (n.d.). *Silent installation of Notepad++*. Notepad++ User Manual. Retrieved December 1, 2025, from <https://npp-user-manual.org/docs/silent-installation/>
- 7-Zip. (n.d.). *7-Zip download*. 7-Zip. Retrieved December 1, 2025, from <https://www.7-zip.org/download.html>
- SourceForge. (n.d.). *7-Zip silent install options*. Retrieved December 1, 2025, from <https://sourceforge.net/p/sevenzips/discussion/45797/thread/>

3) written explanation submitted with your .txt copy of your script.

Line-by-Line Explanation of Script

- `$InstallerPath = "$env:USERPROFILE\Desktop\Installers"` → **Sets the folder on Desktop to save installers.**
- `if (-Not (Test-Path $InstallerPath)) { New-Item -ItemType Directory -Path $InstallerPath | Out-Null }` → **Creates the folder if it doesn't already exist.**
- `$ChromeURL = "https://dl.google.com/dl/chrome/install/googlechromestandaloneenterprise64.msi"` → **Stores download link for Chrome.**
- `$ChromeInstaller = Join-Path $InstallerPath "Chrome.msi"` → **Sets local path for Chrome installer.**
- `Write-Output "Downloading Chrome..."` → **Prints download start message.**
- `Invoke-WebRequest -Uri $ChromeURL -OutFile $ChromeInstaller` → **Downloads Chrome installer.**
- `Write-Output "Downloaded Chrome"` → **Prints download complete message.**
- `Start-Process "msiexec.exe" -ArgumentList "/i \"$ChromeInstaller\" /quiet /norestart"` - `Wait` → **Installs Chrome silently.**
- `Write-Output "Chrome installation complete!"` → **Prints Chrome install complete message.**
- `$NotepadURL = "https://github.com/notepad-plus-plus/notepad-plus-plus/releases/download/v8.6.7/npp.8.6.7.Installer.x64.exe"` → **Stores download link for Notepad++.**
- `$NotepadInstaller = Join-Path $InstallerPath "NotepadPP.exe"` → **Sets local path for Notepad++ installer.**
- `Write-Output "Downloading Notepad++..."` → **Prints download start message.**
- `Invoke-WebRequest -Uri $NotepadURL -OutFile $NotepadInstaller` → **Downloads Notepad++ installer.**
- `Write-Output "Downloaded Notepad++"` → **Prints download complete message.**

- Start-Process \$NotepadInstaller -ArgumentList "/S" -Wait → **Installs Notepad++ silently.**
- Write-Output "Notepad++ installation complete!" → **Prints Notepad++ install complete message.**
- \$SevenZipURL = "https://www.7-zip.org/a/7z2301-x64.exe" → **Stores download link for 7-Zip.**
- \$SevenZipInstaller = Join-Path \$InstallerPath "7zip.exe" → **Sets local path for 7-Zip installer.**
- Write-Output "Downloading 7-Zip..." → **Prints download start message.**
- Invoke-WebRequest -Uri \$SevenZipURL -OutFile \$SevenZipInstaller → **Downloads 7-Zip installer.**
- Write-Output "Downloaded 7-Zip" → **Prints download complete message.**
- Start-Process \$SevenZipInstaller -ArgumentList "/S" -Wait → **Installs 7-Zip silently.**
- Write-Output "7-Zip installation complete!" → **Prints 7-Zip install complete message.**
- Write-Output "All installations complete!" → **Prints final message indicating all installations are done.**

Your submission should include:

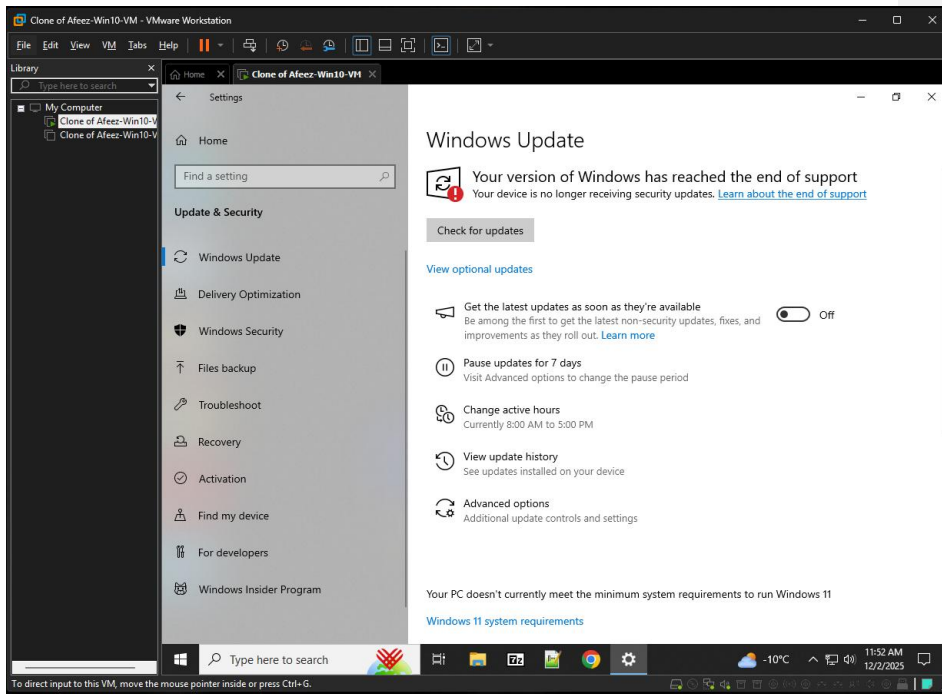
1. Screenshot of your VM with your Computer Name
2. Script for application installation including APA formatted references.
3. If requested, line by line explanation of script.

Part B: Images are Golden (Week 13)

Using VMWare's Clone function you will be creating a "golden image" of a Win 10/11 VM that will be installed by another student on their PC.

Your image must meet the following requirements:

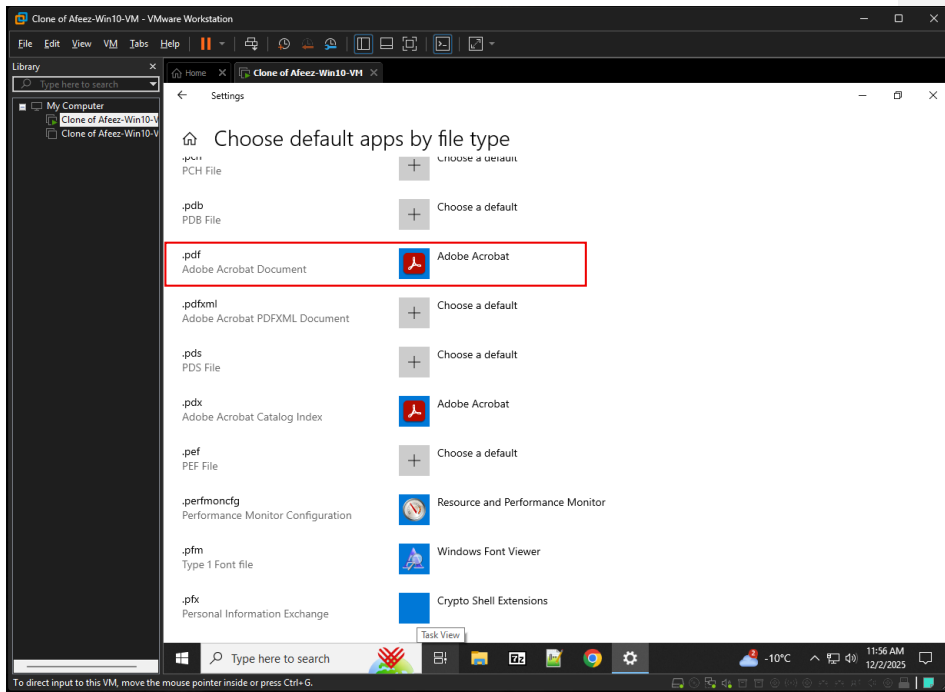
- Windows is fully updated with all the latest patches/security updates



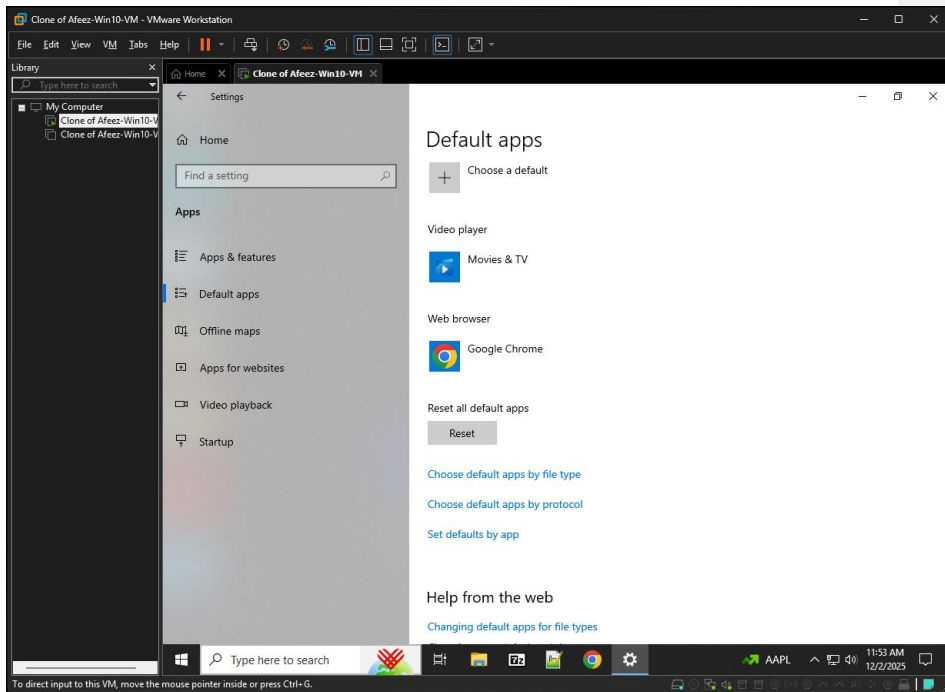
- Your three silently installed applications are present and up to date



- Adobe Reader DC is installed and set as the default PDF application

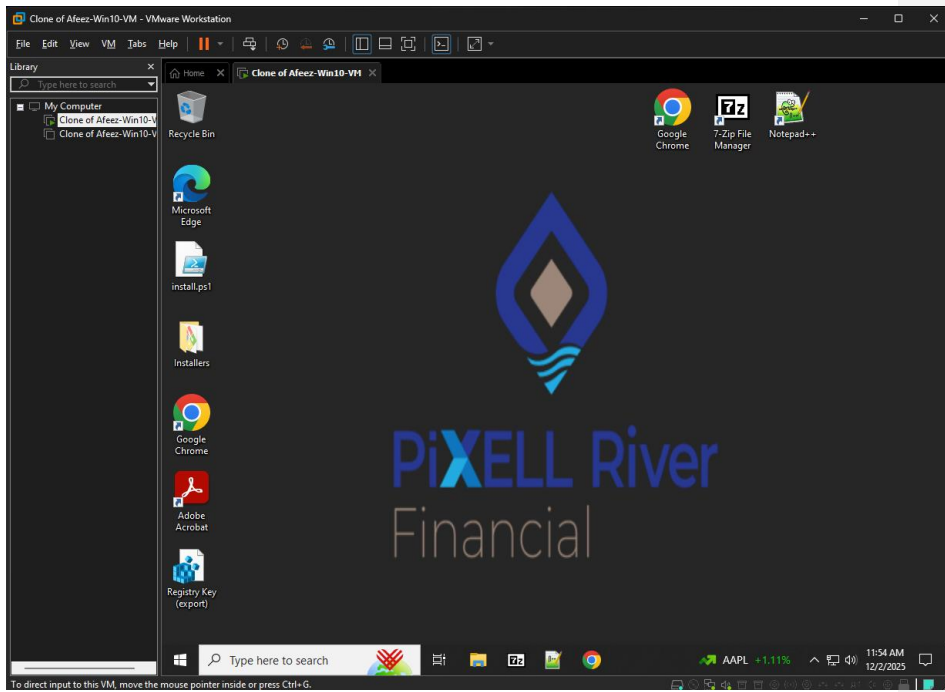


- Firefox or Chrome are installed and set as the default browser

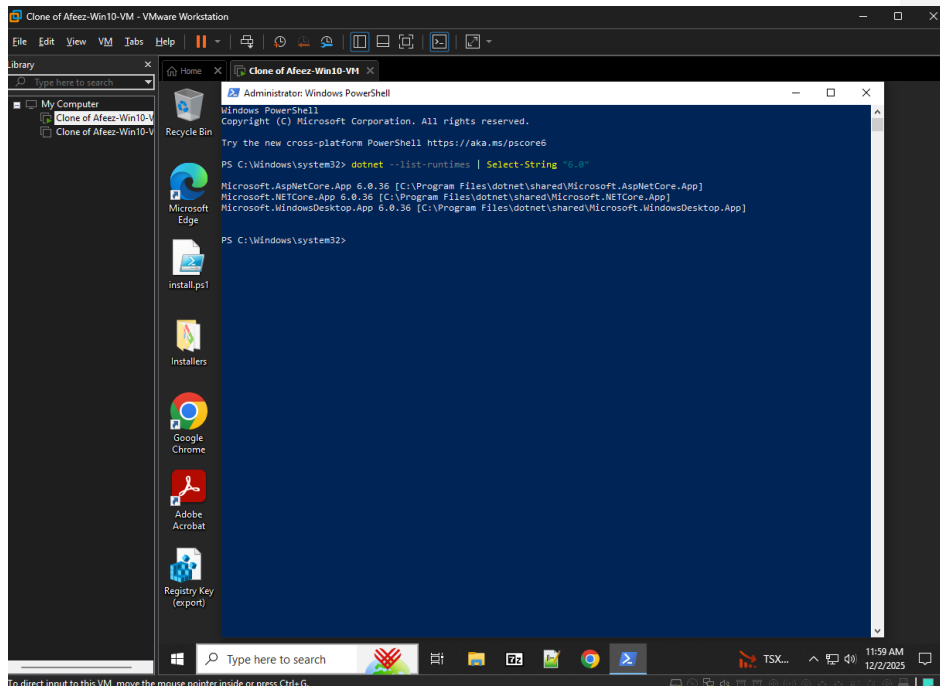


- PiXELL-River Desktop Background has been set

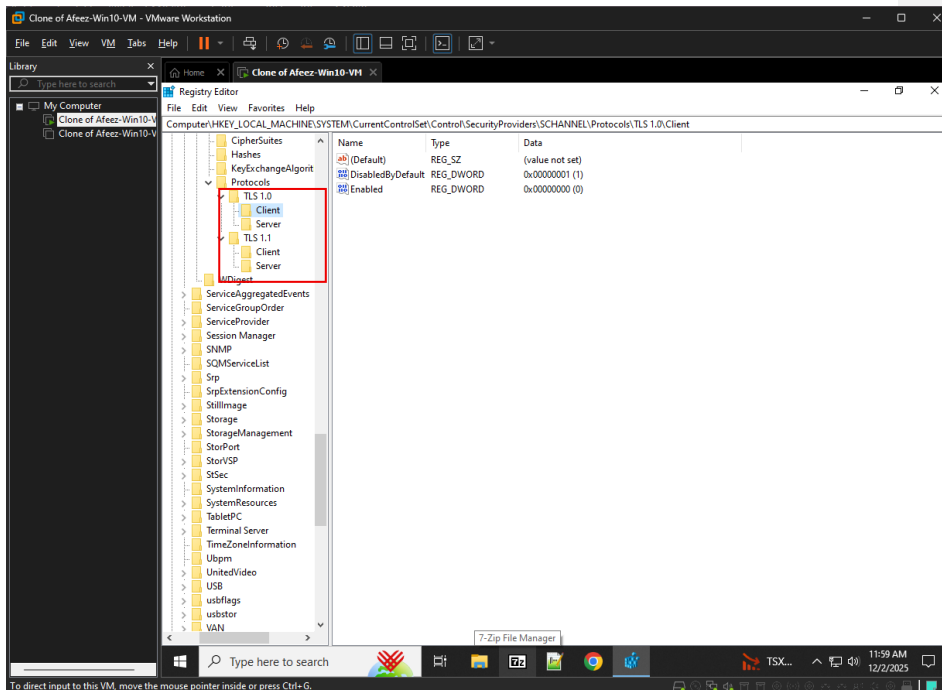
Commented [PS2]: We have this?



- .NET 6.0 has been installed



- Using the registry editor disable TLS 1.0 and 1.1



Your submission should include:

1. Screenshot of your VM with your Computer Name
2. Screenshot of installed apps



3. Export of Registry Key indicating TLS is disabled
Export is attached but this is a screenshot below.-

```
Windows Registry Editor Version 5.00

[HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols]

[HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.0]

[HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.0\Client]
"Enabled"=dword:00000000
"DisabledByDefault"=dword:00000001

[HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.0\Server]
"Enabled"=dword:00000000
"DisabledByDefault"=dword:00000001

[HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.1]

[HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.1\Client]
"DisabledByDefault"=dword:00000001
"Enabled"=dword:00000000

[HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.1\Server]
"DisabledByDefault"=dword:00000001
"Enabled"=dword:00000000
```

4. Screenshot of the .ovf file on your computer, the screenshot should include a time/date stamp. **THIS IS NOT NEEDED IF WE'RE CLONING THE VM INSIDE VMWARE WORKSTATION USING THE VIRTUAL LAB ENVIRONMENT**—
- I used clone option. I sent my .vmx and .vmdk file to ephraim for testing and I tested his golden image.

Commented [PS3]: Too much?

Commented [PS4R3]: Or where you going to get the students to check these items during the checklist part?

Commented [CG5R3]: I was going to have them check it during the checklist evaluation

Part C: Verification in Pairs (Week 14)

Pair up with a partner and swap your exported VM files. You will each import the other person's VM into your VMWare and start it up. Using the provided [checklist](#) evaluate your partner's "golden image" and ensure it meets the requirements. Submit the [checklist](#), and any notes about the evaluation to the dropbox.

Please ensure you check the rubric for the required criteria of this assessment.

Link to Checklist: [Golden Image Checklist.docx](#)

Commented [PS6]: Where is the checklist?

Commented [CG7R6]: It will be in LEARN under this module, but I haven't made it yet

Due Date

Due date can be found in LEARN

Rubric

Rubric can be found in LEARN.